Attraction Diminishing and Distributing for Few-Shot Class-Incremental Learning

Li-Jun Zhao[1], Zhen-Duo Chen[1,*], Yongxin Wang[2], Xin Luo[1], Xin-Shun Xu[1]
[1]School of Software, Shandong University, China
[2]School of Computer Science and Technology, Shandong Jianzhu University, China
lj_zhao1028@163.com, chenzd.sdu@gmail.com, yxinwang@hotmail.com
luoxin.lxin@gmail.com, xuxinshun@sdu.edu.cn

Abstract

Few-Shot Class-Incremental Learning (FSCIL) aims to continuously learn novel classes with limited samples after pre-training on a set of base classes. To avoid catastrophic forgetting and overfitting, most FSCIL methods first train the model on the base classes and then freeze the feature extractor in the incremental sessions. However, the reliance on nearest neighbor classification makes FSCIL prone to the hubness phenomenon, which negatively impacts performance in this dynamic and open scenario. While recent methods attempt to adapt to the dynamic and open nature of FSCIL, they are often limited to biased optimizations to the feature space. In this paper, we pioneer the theoretical analysis of the inherent hubness in FSCIL. To mitigate the negative effects of hubness, we propose a novel Attraction Diminishing and Distributing (D2A) method from the essential perspectives of distance metric and feature space. Extensive experimental results demonstrate that our method can broadly and significantly improve the performance of existing methods.

1. Introduction

Few-Shot Class-Incremental Learning (FSCIL) [1, 5, 6, 9] has been gaining increasing attention in recent years. As a special form of Class Incremental Learning (CIL) [4, 38, 43], FSCIL is proposed to continuously learn novel classes with limited samples after pre-training on base classes with sufficient samples.

In the past few years, FSCIL methods widely adopt the incremental-frozen baseline [17, 32, 37, 45], where the feature extractor is trained only during the base session with base classes. By not adjusting the feature extractor based on the few-shot incremental classes, this baseline fundamentally prevents catastrophic forgetting and overfitting issues that FSCIL focuses on. During the incremental sessions, prototype classifiers are generated based on training samples, and the classification of test samples relies on searching the nearest prototype, i.e., 1-nearest neighbor (NN).

However, the use of NN classification—especially in high-dimensional spaces—is prone to hubness. Specifically, (1) due to neural collapse [21], features from the same class tend to converge towards their within-class mean, while features from different classes are maximally separated. Despite FSCIL aiming to provide a distinct component for each previously seen class in the output distribution, the limited transferability of feature learning often constrains the actual output distribution. Consequently, instead of achieving distinct clusters for all classes, the output feature space is primarily composed of base class clusters. This leads to the output feature distribution dominated by a mixture centered on base class prototypes. (2) Hubness [24] refers to the phenomenon in high-dimensional space where few points are included as the K-NN of other points far more frequently than the remaining data points, and these few points are called hubs. Through analysis, we find that for the output mixture distribution in FSCIL, base class prototypes are generally closer to the mean of the component distributions, and therefore tend to become hubs. This makes the base class prototypes more “attractive” in NN retrieval during inference.

In traditional static and close-set classification, hubness might benefit NN classification by providing more stability in classifications within established class clusters. However, in the dynamic and open-set FSCIL scenario, this skew creates a disadvantage because novel classes are constantly introduced and the hubs from previous classes could dominate the classification of novel data. Some FSCIL methods [28, 33, 40] attempt to adapt to the dynamic and open nature by techniques such as augmentation and fine-tuning. Essentially, they can be viewed as introducing hubs into the feature space and trying to associate them with novel classes.

*Corresponding author

However, without getting to the heart of the problem, these methods are often limited to biased optimizations to the feature space. Not only do these optimizations fail to achieve global optimization, but they may also negatively impact existing local optimizations. Therefore, we step beyond traditional approaches and aim to explore the essence of the problem theoretically.

In this paper, we are the first to theoretically and experimentally analyze the inherent hubness in FSCIL. Based on the analysis, we identify that the hubness is contributed by the distance metric and the positions of sample points in the feature space (i.e., the feature vectors). Therefore, to mitigate the negative influence of certain hubs on other sample points, we propose a D2A method to Diminish the hub "Attraction" from the metric perspective and Distribute the hub "Attraction" from the feature perspective. Specifically, on one hand, a different distance metric is introduced during the incremental sessions to alter the distance metric. This can diminish the hub "attraction" by reducing point correlation. Additionally, we dynamically control the degree of metric alteration based on the current data distribution. On the other hand, in the base session, we modify the loss function to optimize the output distribution. This approach distributes the hub "attraction" by introducing more component distributions. Besides, to minimize negative impacts on the original component distributions, we design a random nearest-neighbor strategy in selecting samples. Our key contributions are summarized as follows:

- We pioneer the analysis of the hubness issue inherent in FSCIL which relies on nearest neighbor classification, especially noting its negative impact in such a dynamic and open setting.
- From the two decisive perspectives of metric and feature, we propose a D2A method to diminish and distribute the hub "attraction".
- Extensive experiments demonstrate that applying our method to existing FSCIL methods can broadly and significantly enhance incremental class accuracy by alleviating hubness, while also boosting overall accuracy.

2. Related Work

Few-Shot Class-Incremental Learning (FSCIL) [2, 15, 18, 44] aims to address the challenge of CIL [10, 27, 31] in scenarios with insufficient data.

Typically, FSCIL methods are not allowed to retain any samples from old classes, and they have fewer incremental class samples compared to CIL. Consequently, during the incremental session, fine-tuning the feature extractor not only faces catastrophic forgetting similar to CIL but also risks overfitting. Such *fine-tuning* methods [14, 40] necessitate strict control over the range and extent of fine-tuning parameters, which not only increases the complexity but also introduces biased optimizations in the feature space.

Given that FSCIL has more base classes compared to CIL, it is feasible to use a similar approach as few-shot learning [23, 29, 35] by learning a unified feature extractor from base classes directly for incremental classes, referred to as the incremental-frozen FSCIL baseline. Therefore, most FSCIL methods follow this baseline, known as *incremental-frozen* methods [13, 19, 32, 37, 45]. However, the fixed feature extractor limits the performance of the incremental classes. Although base classes are crucial for model initialization, the primary challenge and performance focus of the FSCIL paradigm lie in the subsequent incremental classes with insufficient samples. However, since base classes consistently account for a high proportion of all classes, maintaining high accuracy for base classes can maintain high overall accuracy, thereby resulting in low dropping rates. Therefore, it is important to place more emphasis on incremental classes in the evaluation metrics.

While some recent methods [7, 22, 36, 40] have acknowledged the low accuracy of incremental classes and attempted to address this issue, they attribute the cause to a single semantic or feature-based reason. Some of these methods [12, 32] purposefully adjust prototype classifiers during the incremental sessions, which essentially lead to misalignment between the feature extractor and prototype classifiers. Some methods [3, 34] preset sufficient targets for all upcoming classes to be optimized, but since FSCIL is an open scenario, most methods adhere to the principle that the number of incremental classes should ideally remain unrestricted (i.e., potentially infinite). In contrast, we systematically analyze the underlying causes from a theoretical perspective and propose a solution from the perspectives of distance metric and feature space, avoiding direct alterations to the positions of prototype classifiers and restrictions on openness.

3. Methodology

3.1. Task Definition

In FSCIL, the model is trained on a sequence of training set $\{\mathcal{D}_{train}^t\}_{t=0}^T$. Each $\mathcal{D}_{train}^t$ comprises a set of pairs $(\mathbf{x}_i, y_i)$ from session t, where $\mathbf{x}_i$ is a sample from class $y_i \in \mathcal{C}^t$. $\mathcal{C}^t$ is the label set of dataset $\mathcal{D}_{train}^t$, and $|\mathcal{C}^t|$ represents the number of class labels in the set $\mathcal{C}^t$. The training set $\mathcal{D}_{train}^0$ contains sufficient samples for a large number of classes. The subsequent training set $\mathcal{D}_{train}^t (t > 0)$ with limited samples can be organized as N-way K-shot format, meaning that there are only K samples for each of the N classes from $\mathcal{C}^t$. Session 0 is called **base session**, and session $t (t > 0)$ is called **incremental session**. During each session t, the model can only access $\mathcal{D}_{train}^t$. However, it needs to be tested on samples from all seen classes, i.e., $\mathcal{C}^0 \cup \mathcal{C}^1 \cdots \cup \mathcal{C}^t$. The classes from $\mathcal{C}^0$ are referred to as **base classes**, while those from $\mathcal{C}^1 \cdots \cup \mathcal{C}^t$ are termed **incremen-**

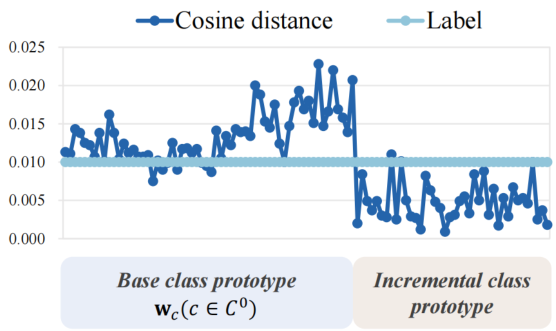


Figure 1. Visualization of $N_1^{n,d}(\mathbf{w}_c)/n$ on the *mini*ImageNet dataset in the last session, where $n = 10000$ and $d = 512$. According to the class labels, the number of nearest neighbors should be evenly distributed across the 100 classes, implying $N_1^{n,d}(\mathbf{w}_c)/n = 0.01$ for all class prototypes.

tal classes.

Specifically, the training objective of the base session is to enable the model to recognize all base classes $\mathcal{C}^0$ by minimizing the classification loss function,

$$\mathcal{L}_{base} = \frac{1}{|\mathcal{D}^0_{train}|} \sum_{(\mathbf{x}_i, y_i) \in \mathcal{D}^0_{train}} \mathcal{L}(\phi(\mathbf{x}_i), y_i), \tag{1}$$

where model $\phi(\cdot)$ can be decomposed into feature extractor $f(\cdot)$ and linear classifiers $\mathbf{W}$, i.e., $\phi(\mathbf{x}) = \mathbf{W}^\top f(\mathbf{x}) \in \mathbb{R}^{|\mathcal{C}^0| \times 1}$, where $f(\mathbf{x}) \in \mathbb{R}^d$ and $\mathbf{W} = \{\mathbf{w'}_c | c \in \mathcal{C}^0\} \in \mathbb{R}^{d \times |\mathcal{C}^0|}$. Typically, $f(\mathbf{x})$ and $\mathbf{w'}_c$ are L2 normalized, so the output logit $\phi(\mathbf{x})$ denotes cosine similarity between $f(\mathbf{x})$ and each $\mathbf{w'}_c$. In the incremental session t, the weight of the classifier $\mathbf{W}$ needs to be expanded for novel classes, i.e., $\mathbf{W} = \{\mathbf{w'}_c | c \in \cup_{j=0}^t \mathcal{C}^j\} \in \mathbb{R}^{d \times \sum_{j=0}^t |\mathcal{C}^j|}$.

A widely adopted solution (incremental-frozen baseline) [37] is to freeze feature extractor $f(\cdot)$ in the incremental sessions to avoid the catastrophic forgetting and overfitting problems that FSCIL methods focus on. Therefore, for i.i.d. input data, the expected output distribution of the feature extractor will remain consistent with the distribution established in the base session. Besides, the mean vector of all training samples (i.e., the prototype) of each class c will be used as the classifier's weight for this class in the incremental sessions, i.e., $\mathbf{w}_c = \frac{1}{Num_c} \sum_{y_i = c} f(\mathbf{x}_i)$, where Num_c indicates the number of training samples in class c. During the inference phase, nearest-neighbor prototype classifiers are employed.

3.2. Hubness in FSCIL

In this section, to demonstrate the existence of hubness in FSCIL, we first define the feature space output distribution obtained during the base training phase based on neural collapse. Building upon this, we analyze the conditions for the formation of hubness in relation to distance metrics. Ultimately, we prove that base class prototypes tend to become hubs.

Definition 3.1 (Neural collapse) *Neural Collapse refers to a phenomenon observed in classification problems during the final stages of training, where the features* $\mathbf{a}$ *extracted by the feature extractor for samples of the same class converge to their within-class mean, i.e., covariance* $\mathbf{\Sigma}_c \to \mathbf{0}$ *and* $\mu_c \to \mathbf{w'}_c$*, and features for different classes become maximally separated in the feature space.*

Given the above properties, in FSCIL, after base training under Eq. (1), the output from the feature extractor $f(\mathbf{x}) \in \mathbb{R}^d$ for i.i.d. samples can be expressed as a mixture distribution,

$$\mathcal{P}(\mathbf{a}) = \sum_{c=1}^{|\mathcal{C}^0|+\delta} \pi_c \mathbf{N}(\mathbf{a}; \mu_c, \mathbf{\Sigma}_c), \tag{2}$$

where $|\mathcal{C}^0| + \delta$ is the number of components in the mixture distribution, π_c is the mixing coefficient for the c-th component, and $\mathbf{N}$ represents a normal distribution. The class prototype $\mathbf{w}_c$ converges to μ_c and $\delta \to 0$. A detailed proof of the output distribution is provided in the supplementary materials. This represents the most challenging case where the feature extractor, trained solely on the base classes, fails to extract discriminative features for the incremental classes to form clusters. The opposite simplest case (i.e., $\delta = \sum_{j=1}^t |\mathcal{C}^j|$) requires no further analysis.

Consider a single component, let $\mathbf{a}_i = [a_i^1, ..., a_i^d]^T, i = 0, ..., n$ be a sample of $n+1$ i.i.d. points from $\mathbf{N}(\mathbf{a}; \mu_c, \mathbf{\Sigma}_c)$. Assume that $\mathbf{N}$ factorizes as $\mathbf{N}(\mathbf{a}; \mu_c, \mathbf{\Sigma}_c) = \prod_{l=1}^d \mathcal{N}(a^l; \mu_c, \mathbf{\Sigma}_c)$, i.e., the coordinates $a^1, ..., a^d$ are i.i.d. The general simplified form of the distance metric between two points is defined as $\mathcal{E}(\mathbf{a}_i, \mathbf{a}_j) = \sum_{l=1}^d g(a_i^l, a_j^l)$. Since the optimization objective in FSCIL during training is in the form of cosine similarity, the $\mathcal{E}$ here is the corresponding distance form, i.e., cosine distance $1 - \cos(\mathbf{a}_i, \mathbf{a}_j)$.

Theorem 3.2 *Let* $N_1^{n,d}(\mathbf{a})$ *denotes the number of points among other* n *points whose nearest neighbor is* $\mathbf{a}$*. Set* $\beta = \mathrm{Correlation}(g(X,Y), g(X,Z))$*, where* X, Y, Z *are i.i.d vector component variables with common distribution* $\mathcal{N}$*.*

If $\beta = 0$ *then*

$$\lim_{n\to\infty} \lim_{d\to\infty} \mathrm{Var}(N_1^{n,d}) = 1, \tag{3}$$

$$\lim_{n\to\infty} \lim_{d\to\infty} N_1^{n,d} = \mathit{Poisson}(\lambda = 1). \tag{4}$$

If $\beta > 0$ *then*

$$\lim_{n\to\infty} \lim_{d\to\infty} \mathrm{Var}(N_1^{n,d}) = \infty, \lim_{n\to\infty} \lim_{d\to\infty} N_1^{n,d} = 0. \tag{5}$$

Further details related to the theorem can be found in [24, 25]. In FSCIL, the correlation between $g(X,Y)$ and $g(X,Z)$ with respect to the dimension is greater than 0 under cosine distance, i.e., $\beta > 0$. It indicates that the number of nearest neighbors is concentrated on a small subset of points in high-dimensional space, indicating the existence of hubness, where these few points are referred to as hubs.

Then, we prove that in the FSCIL problem, base class prototypes tend to become hubs. Specifically, consider two points $\mathbf{a}_1$ and $\mathbf{a}_2$ randomly sampled from $\mathbf{N}(\mathbf{a}; \mu_c, \boldsymbol{\Sigma}_c)$, s.t.,

$$\mathcal{E}(\mathbf{a}_1, \mu_c) - \mathcal{E}(\mathbf{a}_2, \mu_c) = \cos(\mathbf{a}_2, \mu_c) - \cos(\mathbf{a}_1, \mu_c) = \gamma s, \tag{6}$$

where γ is a constant, $s = \sqrt{\mathrm{Var}(\mathcal{E}(\mathbf{a}, \mu_c))}$. For $\hat{\mathbf{a}}$ also sampled from $\mathcal{N}(\mathbf{a}; \mu_c, \Sigma_c)$, we have the expected difference

$$\begin{aligned}\Delta &= \mathbb{E}(1 - \cos(\mathbf{a}_1, \hat{\mathbf{a}})) - \mathbb{E}(1 - \cos(\mathbf{a}_2, \hat{\mathbf{a}})) \\ &= -[\cos(\mathbf{a}_1, \mathbb{E}(\hat{\mathbf{a}})) - \cos(\mathbf{a}_2, \mathbb{E}(\hat{\mathbf{a}}))] \\ &= \gamma s.\end{aligned} \tag{7}$$

Therefore, a random sample $\hat{\mathbf{a}}$ is expected to be closer to $\mathbf{a}_i$ if $\mathbf{a}_i$ is closer to distribution mean μ_c.

Based on the analysis from Definition 3.1, base class prototypes $\mathbf{w}_c$ close to the cluster mean μ_c under the cosine **distance metric** $\mathcal{E}$, so they tend to be hubs in the **feature space** $f(\mathbf{x})$. Furthermore, in Fig. 1, we visualize $N_1^{n,d}(\mathbf{w}_c)/n$ for all prototypes among n test samples based on the baseline method [29]. It can be observed that under the cosine distance metric, $N_1^{n,d}(\mathbf{w}_c)/n (c \in \mathcal{C}^0)$ is generally larger than the value obtained by labels. Therefore, the more "popular" and "attractive" base class prototypes severely disrupt the accuracy of test samples from incremental classes (Fig. 2).

Given that both distance metric and feature space contribute to this phenomenon, we design a method from both perspectives—metric (Sec. 3.3) and feature (Sec. 3.4)—to alleviate it, thereby improving performance on the FSCIL task.

3.3. Diminishing Attraction

From a metric perspective, based on Theorem 3.2, *for i.i.d* X, Y, Z, *if* $\beta = 0$, *the number of nearest neighbors* $N_1^{n,d}$ *for each point follows a Poisson distribution, meaning that all points are uniformly distributed over* $\mathbb{R}^d$. Therefore, altering distance metric $\mathcal{E}(\mathbf{a}_i, \mathbf{a}_j) = \sum_{l=1}^{d} g(a_i^l, a_j^l)$ within the existing feature space can reduce point correlation β, thereby diminishing the "attraction" of base prototypes in the incremental sessions.

During the inference phase of the typical FSCIL setup, the feature extractor $f(\cdot)$ and prototype classifiers $\mathbf{W}$ are utilized to predict the class of a sample. In session t, for a given test sample $\mathbf{x}_i$, the prediction result is determined by

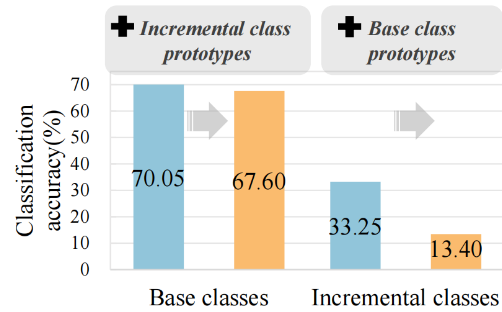


Figure 2. Visualization of accuracy on the *mini*ImageNet dataset in the last session. The left/right side shows the impact on the accuracy of adding incremental/base class prototypes when using base/incremental class prototypes for NN classification on base/incremental class test samples.

taking the argmin over a set of cosine distances (or argmax over a set of cosine similarities),

$$y_i^\star = \underset{c \in \bigcup_{j=0}^{t} \mathcal{C}^j}{\mathrm{argmin}}\ \mathcal{E}(\mathbf{w}_c, \mathbf{f}_i) = \underset{c \in \bigcup_{j=0}^{t} \mathcal{C}^j}{\mathrm{argmin}} \sum_{l=1}^{d} g(w_c^l, f_i^l), \tag{8}$$

$$g(w_c^l, f_i^l) = \frac{1}{d} - \frac{w_c^l \cdot f_i^l}{\|\mathbf{w}_c\| \cdot \|\mathbf{f}_i\|}, \tag{9}$$

where $\mathbf{f}_i$ is the abbreviation for $f(\mathbf{x}_i)$, and $\|\cdot\|$ represents the L2 norm of the vector.

The above describes the original inference process, where the distance metric $\mathcal{E}$ used aligns with the training objective. To maintain the similarity relationships learned during training while reducing β, we can introduce a distance metric $\mathcal{E}'$ that differs from $\mathcal{E}$ but remains positively correlated with it. A completely unrelated metric would severely impair classification accuracy. Here, we use the Manhattan distance as an example,

$$\mathcal{E}'(\mathbf{w}_c, \mathbf{f}_i) = \sum_{l=1}^{d} g'(w_c^l, f_i^l) = \sum_{l=1}^{d} |w_c^l - f_i^l|. \tag{10}$$

Subsequently, we merge the distance metrics $\mathcal{E}$ and $\mathcal{E}'$ into an overall metric $\mathcal{E}^{overall}$. To eliminate their disparity in scale, we first scale $\mathcal{E}$ and $\mathcal{E}'$ to distributions with the mean of 1, then apply a weighted summation to them:

$$\begin{aligned}\mathcal{E}_{c,i}^{overall} =& (1-\gamma) \frac{\mathcal{E}_{c,i}}{\sum_{c \in \bigcup_{j=0}^{t} \mathcal{C}^j} \mathcal{E}_{c,i} / \sum_{j=0}^{t} |\mathcal{C}^j|} \\ &+ \gamma \frac{\mathcal{E}'_{c,i}}{\sum_{c \in \bigcup_{j=0}^{t} \mathcal{C}^j} \mathcal{E}'_{c,i} / \sum_{j=0}^{t} |\mathcal{C}^j|},\end{aligned} \tag{11}$$

where γ represents the degree of metric alteration and $\mathcal{E}_{c,i}$ is the abbreviation for $\mathcal{E}(\mathbf{w}_c, \mathbf{f}_i)$. The scale disparity between different metrics is eliminated by dividing each metric by its average value. Eq. (11) is exemplified with a single test sample $\mathbf{x}_i$, so the denominator is the average over the single sample. In practical applications where n test samples may be processed simultaneously, the denominator can be directly replaced with the average across all test samples. Therefore, in the incremental sessions, the prediction process of Eq. (8) becomes

$$y_i^\star = \underset{c \in \cup_{j=0}^{t} \mathcal{C}^j}{\text{argmin}} \ \mathcal{E}_{c,i}^{overall}. \tag{12}$$

Since the closer the incremental class distribution is to the base class distribution, the larger the required value of γ, we design γ to adaptively adjust according to the data distribution in each session t,

$$\gamma = \frac{\tanh(\eta - \mathrm{M}_D^t) + 1}{2}, \tag{13}$$

where $\tanh$ is used to smoothly map values to the range $[-1, 1]$, η is a fixed central axis hyperparameter, and M_D^t represents the distance between incremental class distribution and base class distribution in session t. Specifically, M_D^t is obtained by calculating the Maximum Mean Discrepancy (MMD) [11] between base class prototypes and incremental class prototypes,

$$\mathrm{M}_D^t = \left\| \frac{1}{|\mathcal{C}^0|} \sum_{c \in \mathcal{C}^0} h(\mathbf{w}_c) - \frac{1}{\sum_{j=1}^{t} |\mathcal{C}^j|} \sum_{c \in \cup_{j=1}^{t} \mathcal{C}^j} h(\mathbf{w}_c) \right\|_{\mathcal{H}} \tag{14}$$

where h maps the feature distribution to reproducing kernel hilbert space and $\|\cdot\|_{\mathcal{H}}$ represents the norm in this space.

3.4. Distributing Attraction

From a feature perspective, *in a mixture of unimodal distributions, hubs tend to appear near the means of individual component distributions*. Besides, since classification still depends on the similarity relationships learned during training, it is not feasible to alter the metric to the extent that the attraction of the base class prototypes is completely eliminated. Based on the above analysis, we can optimize a mixture distribution with more components δ during training, thus further distributing the "attraction" of base prototypes in the incremental sessions.

To optimize the output distribution to conclude more component distributions, the classification loss function Eq. (1) during the base training phase is modified as follows,

$$\mathcal{L}_{overall} = (1 - \alpha)\mathcal{L}_{base} + \alpha \mathcal{L}_{gap}, \tag{15}$$

where α represents the optimization strength for the additional component distributions. The loss function $\mathcal{L}_{gap}$ is designed to encourage gap samples $\mathbf{x}_{ij}$ between the original component distributions of class y_i and class y_j to form clusters. Specifically, it is formulated as follows,

$$\mathcal{L}_{gap} = \frac{1}{|\mathcal{D}_{gap}|} \sum_{(\mathbf{x}_{ij}, y_i, y_j) \in \mathcal{D}_{gap}} \frac{\mathcal{L}(\phi(\mathbf{x}_{ij}), y_i) + \mathcal{L}(\phi(\mathbf{x}_{ij}), y_j)}{2}. \tag{16}$$

To minimize negative impacts on the original component distributions and the already filled component distributions, we design a random nearest-neighbor strategy to construct the gap dataset $\mathcal{D}_{gap}$, utilizing the nearest-neighbor boundary samples as described in Algorithm 1. By incorporating randomness, this approach avoids excessive repetition in nearest-neighbor selection, thereby increasing the diversity of the selected samples and promoting the formation of more component distributions.

Algorithm 1 Random nearest-neighbor strategy.

Input: A batch of training samples $\mathcal{B} = \{\mathbf{x}_i \mid i = 0, \ldots, n\}$, labels y_i for each $\mathbf{x}_i$, number of samples r

Output: Gap dataset $\mathcal{D}_{\text{gap}}$

1: **for** each sample $\mathbf{x}_i \in \mathcal{B}$ **do**

2: Compute the cosine similarity $\cos(\mathbf{x}_i, \mathbf{x}_j)$ with all other samples $\mathbf{x}_j \in \mathcal{B}$ where $y_j \neq y_i$.

3: Calculate the probability of selecting sample $\mathbf{x}_{j'}$ based on cosine similarity:

$$p_{ij'} = \frac{\cos(\mathbf{x}_i, \mathbf{x}_{j'}) + 1}{\sum_{j | y_j \neq y_i} (\cos(\mathbf{x}_i, \mathbf{x}_j) + 1)} \tag{17}$$

// The addition of 1 ensures all probabilities are positive.

4: Sample r instances without replacement from the multinomial distribution with probabilities p_{ij}:

$$\mathcal{I}_i \sim \text{Mult}(r; \{p_{ij} \mid \mathbf{x}_j \in \mathcal{B}, y_j \neq y_i\}) \tag{18}$$

// The samples closer to $\mathbf{x}_i$ are more likely to be selected.

5: Compute the gap samples $\{\mathbf{x}_{ij} = \frac{1}{2}(\mathbf{x}_i + \mathbf{x}_j) | \mathbf{x}_j \in \mathcal{I}_i\}$

6: **end for**

7: Construct the gap dataset $\mathcal{D}_{\text{gap}} = \{(\mathbf{x}_{ij}, y_i, y_j) | \mathbf{x}_j \in \mathcal{I}_i, i \in \mathcal{B}\}$.

Dataset	$\lvert\mathcal{C}^0\rvert$	T	N	K
*mini*ImageNet [26]	60	8	5	5
CIFAR100 [16]	60	8	5	5
CUB200 [30]	100	10	10	5

Table 1. Statistics of datasets.

Method	Incremental class accuracy in each session (%)								Overall accuracy (%)
	1	2	3	4	5	6	7	8	Last (8)
FACT [41]	15.80	14.40	15.40	14.55	13.64	12.20	12.66	13.20	48.51
C-FSCIL [12]	5.20	12.40	17.27	19.90	23.40	22.60	25.91	25.95	51.41
TEEN [32]	40.20	35.60	32.47	32.70	29.96	28.33	28.89	29.35	52.08
Bidist [40]	27.00	30.30	29.60	27.85	28.44	26.80	26.23	25.62	52.22
OSHHG [7]	39.26	32.26	31.31	30.12	26.95	24.08	24.51	25.89	46.75
EHS [8]	-	-	-	-	-	-	-	-	49.06
DyCR [20]	14.40	13.80	12.93	13.95	12.12	11.20	12.31	13.90	51.08
CLOSER [19]	30.20	28.70	27.93	26.90	25.16	23.67	24.26	25.78	53.33
M2SD [17]	-	-	-	-	-	-	-	-	56.51
Decoupled-Cosine [29]	11.20	12.70	13.87	14.10	13.36	12.43	13.09	13.40	45.92
+D2A (Ours)	15.20	18.50	20.13	20.75	19.48	18.63	19.31	20.50	49.33
CEC [37]	15.40	17.10	16.67	15.90	14.20	13.63	14.20	14.78	47.63
+D2A (Ours)	20.80	21.70	22.67	22.60	20.64	19.77	20.43	21.30	51.34
CLOM [45]	13.00	14.20	14.07	14.00	12.88	12.20	13.26	15.05	47.95
+D2A (Ours)	15.00	18.00	19.73	21.35	21.20	19.70	19.94	21.33	48.54
SAVC [28]	38.00	33.90	30.80	30.60	26.68	24.80	26.23	27.13	55.97
+D2A (Ours)	**46.00**	**40.20**	**38.40**	**40.05**	**36.12**	**33.83**	**34.17**	**36.33**	**57.72**

Table 2. The incremental class accuracy of each incremental session and the overall accuracy of the last session on the *mini*ImageNet dataset.

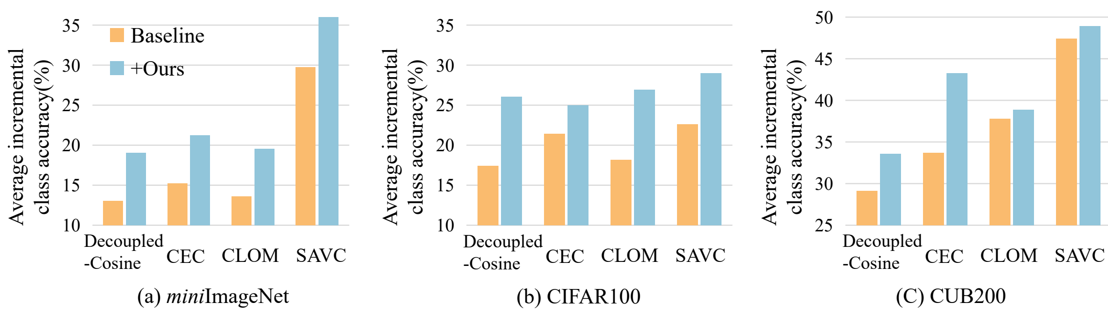


Figure 3. The average incremental class accuracy across different sessions on the *mini*ImageNet, CIFAR100, and CUB200 datasets.

4. Experiments

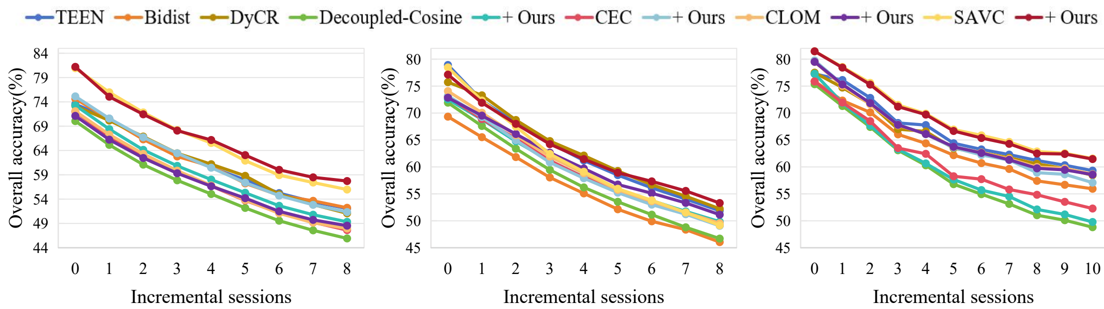
4.1. Experimental Setup

4.1.1. Datasets

Following the benchmark setting [37], we conduct experiments on *mini*ImageNet [26], CIFAR100 [16], and CUB200 [30] datasets. The statistical characteristics of the three datasets are listed in Table 1.

4.1.2. Experimental details

For a fair comparison, training-related settings are consistent with the baseline methods, and results for all baseline methods are reproduced using publicly available code under the same experimental environment. Besides, the results of the comparative methods that are not reported in their papers are reproduced by their publicly available source code. We set α as 0.5 to balance the optimization strength between the original and additional components across all datasets and baselines. r is set as 4 to ensure diversity in sample pairs and kept small for efficiency for all datasets and baselines. To enable γ to dynamically adjust around 0.5 based on session distribution, η is fixed as a decimal near M_D^1. Specifically, for Decoupled-Cosine, CEC, and CLOM, η is set to 0.1 for CIFAR100 and 0.3 for *mini*ImageNet and CUB200, while for SAVC, η is set to 0.2 across all datasets. Since CUB200 is a fine-grained dataset with subtle inter-class differences, we do not apply the Distributing Attraction to baseline methods with complex feature extraction processes (i.e., CLOM and SAVC) on the CUB200 dataset.



(a) *mini*ImageNet (b) CIFAR100 (C) CUB200

Figure 4. The overall accuracy of each session on the *mini*ImageNet, CIFAR100, and CUB200 datasets.

IM	TRN	IA	1		3		5		7		Last (8)	
			Inc.	Overall	Inc.	Overall	Inc.	Overall	Inc.	Overall	Inc.	Overall
			22.20	67.60	17.33	59.45	16.80	53.53	15.71	48.80	15.45	46.70
√			28.00	67.60	22.47	59.53	21.80	53.96	19.86	49.24	19.75	47.26
	√		28.60	68.38	22.00	60.67	22.16	55.02	22.71	51.19	22.18	49.27
√	√		**31.20**	68.08	**25.00**	60.59	**24.96**	55.34	**25.29**	51.67	**24.68**	49.81
√	√	√	30.08	**68.82**	24.93	**61.11**	**24.96**	**55.59**	**25.29**	**51.74**	**24.68**	**49.84**

Table 3. Ablation studies of our proposed method on the CIFAR100 dataset. Due to space limitations, only the Incremental class accuracy and overall accuracy on sessions 1, 3, 5, 7, and 8 are presented.

4.2. Comparison to Existing Methods

In this section, we comprehensively apply our method to existing FSCIL methods for performance comparison and evaluate it against other representative and recent methods.

Applying our method to the incremental-frozen FSCIL methods yields substantial performance improvements. As shown in Tab. 2 and Fig. 3, by alleviating hubness attraction, our method significantly enhances the accuracy of challenging incremental classes and achieves state-of-the-art performance. Although SAVC [28] is already dedicated to adapting incremental classes, our method still notably boosts its performance, surpassing existing methods that specifically consider incremental class accuracy, such as TEEN [32] and OSHHG [7]. Additionally, Tab. 2 and Fig. 4 show that incorporating our method leads to a general improvement in overall accuracy and surpasses recent methods. As the number of incremental classes rises, the advantage of our method becomes even more evident, indicating that our approach provides genuine, balanced FSCIL performance improvements, rather than simply preserving base class performance statically. Additional detailed comparative results are provided in the supplementary materials.

4.3. Ablation Study

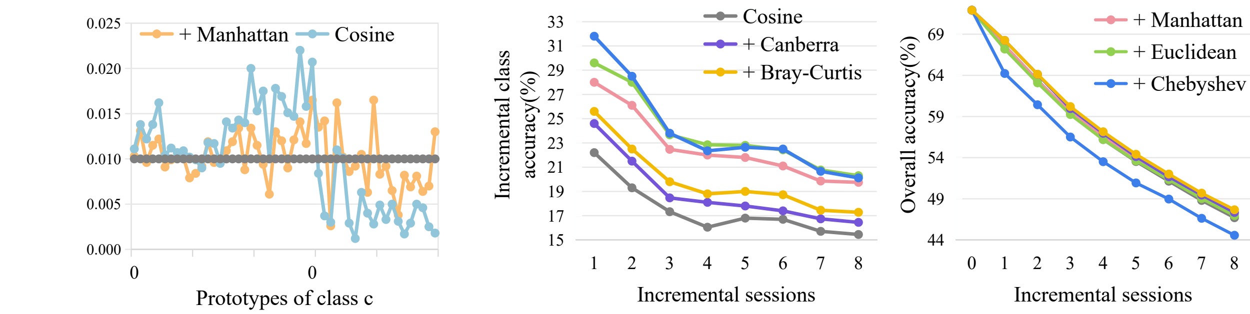
To demonstrate the effectiveness of each component in our method, we conduct ablation experiments based on the baseline method [29] in Tab. 3.

First, in the incremental sessions, altering the distance metric during inference (**IM**) as described in Eq. (11) ($\gamma = 0.5$) not only significantly improves the accuracy of incremental classes but also boosts overall accuracy. This demonstrates that **IM** can diminish the "attraction" of base prototypes and lead to performance improvements. Further experimental analysis in Sec. 4.4.1 explains its deep mechanism and general effectiveness. Based on this, altering the feature distribution during base training (**TRN**) as described in Sec. 3.4 can further improve performance. Further experimental analysis in Sec. 4.4.2 demonstrates the long-term effectiveness of the random nearest-neighbor strategy. Finally, dynamically adjusting the degree of metric alteration during inference (**IA**) as described in Eq. (13) can better maintain overall accuracy.

4.4. Further Analysis

4.4.1. Delve into Metrics

To validate that altering distance metrics can decrease hubness attraction, we visualize the changes in the values of $N_1^{n,d}(\mathbf{w}_c)/n$ after introducing Manhattan distance on the *mini*ImageNet dataset and the baseline method [29] in the last session, as shown in Fig. 5 (a), where n represents the total number of test samples. It can be observed that $N_1^{n,d}(\mathbf{w}_c)/n$ for base class prototypes generally decreases, narrowing the gap in attraction between the base class pro-



(a) Effect of metric alteration on $N_1^{n,d}(\mathbf{w}_c)/n$ (b) Impact of introducing different types of distance metrics on performance

Figure 5. Analysis of altering distance metrics based on Eq. (11), where $\gamma = 0.5$.

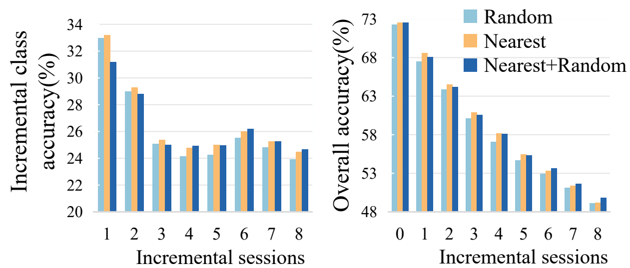


Figure 6. Analysis of different sample strategies to construct the gap dataset.

totypes and incremental class prototypes.

Furthermore, we compare the impact of introducing various distance metrics on performance on the CIFAR100 dataset, as shown in Fig. 5 (b), and observe general performance improvements.

Specifically, cosine distance normalizes vectors, mitigating dimension-specific differences by focusing on vector direction rather than magnitude. This allows the training process to emphasize relationships that are robust to scale variations across dimensions. Manhattan ($p = 1$), Euclidean ($p = 2$), and Chebyshev ($p \to \infty$) distances, as special cases of L^p distances, increasingly emphasize dimensions with larger absolute differences as p increases. While the Chebyshev distance's greater alteration relative to cosine distance significantly enhances incremental class accuracy, it also excessively distorts similarity relationships. Metrics like Bray-Curtis and Canberra, which also reduce the impact of dimension-specific differences, preserve the cosine-based relationships learned in training, helping maintain or even improve overall accuracy. However, due to their limited degree of alteration, these metrics yield moderate improvements in incremental class performance.

In summary, our method is effective with most distance metrics. We adopt the Manhattan distance because it not only provides a pronounced improvement in incremental class accuracy but also boosts overall accuracy.

4.4.2. Different Sample Strategy

To demonstrate the advantages of the random nearest-neighbor strategy, we conduct additional experiments on the CIFAR100 dataset in Fig. 6.

It can be observed that consistently selecting the nearest samples (**Nearest**) yields higher accuracy compared to completely random selection (**Random**) in the initial few sessions. This indicates that the choice of nearest neighbors minimizes the negative impact on the original component distributions. However, in the later sessions, further introducing randomness (**Nearest+Random**) leads to optimal performance in terms of both overall accuracy and incremental class accuracy. This suggests that as new classes continue to arrive, methods with a stronger ability to introduce more component distributions can achieve superior performance. Since sustained performance is more important in the FSCIL task, we adopt the random nearest-neighbor strategy.

5. Conclusion

In conclusion, Few-Shot Class-Incremental Learning (FSCIL) faces significant challenges due to the hubness phenomenon, which arises from the reliance on nearest neighbor (NN) classification. This phenomenon negatively affects performance in the dynamic and open FSCIL task. While recent methods attempt to address the dynamic and open nature of FSCIL, they are often limited by biased optimizations to the feature space. In this paper, we have pioneered the theoretical analysis of the inherent hubness in FSCIL and proposed a novel method, Attraction Diminishing and Distributing (D2A). By considering both distance metrics and feature space, our method effectively reduces the negative impact of hubness. Extensive experimental results validate the effectiveness of our method, demonstrating significant improvements in incremental class accuracy and overall accuracy.

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